

EVS Assignment 2

Course: B.Tech. Computer Science & Engineering- IV Sem. II year

Institution: School of Engineering & Technology, CSJM University Kanpur

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1. What actions are you going to take to play the above role in Environment?

Answer: > As a software developer, my primary action will be to design and implement highly optimized, resource-efficient algorithms that minimize computational overhead, directly reducing the carbon footprint and energy consumption of data centers. I intend to leverage Green Computing practices, cloud optimization, and smart automated energy-monitoring tools to track and curb unnecessary electricity usage. Additionally, I will focus on building digital solutions, such as IoT-enabled monitoring applications, that can track waste, optimize water supply distribution, and help communities manage local natural resources more sustainably through real-time data transparency.

2. Define your societal responsibility as an engineer in the management of Natural Resources to eradicate the poverty.

Answer: > My societal responsibility is to build equitable, open-source technology that democratizes access to resource management data for underserved communities. Inefficient resource allocation deeply impacts those below the poverty line. By developing data-driven platforms and predictive analytics models, we can optimize agricultural supply chains, forecast crop yields, and ensure fair distribution of water and clean energy. Providing small-scale farmers and local administrators with accessible, low-bandwidth mobile applications allows them to prevent resource waste, maximize their economic output, and use technology as a direct lever to break the cycle of poverty.

3. Define your actions to play your role as an engineer, that you can remember as Lifelong learning from this institute (which you like most).

Answer: > The most vital lifelong lesson I will carry forward from this institute is the mindset of sustainable engineering—understanding that the code or systems we build do not exist in a vacuum, but have real-world environmental impacts. My action plan involves continuous upskilling in green software engineering practices and emerging frameworks that prioritize energy efficiency. I commit to participating in collaborative technical forums, open-source sustainability initiatives, and community mentorship programs, ensuring that my engineering knowledge evolves to solve pressing societal challenges long after graduation.



- **Title of Image:**

Harmonizing Nature and Human Development: A Call for Smart Resource Management

- **Legends of Image:**

This image captures a fragile mountain river ecosystem running directly alongside human settlement and commercial infrastructure. It perfectly illustrates the intersection where engineering responsibilities must step in. As a Computer Science engineer, looking at this scene highlights the urgent need for data-driven environmental tracking. Unregulated construction and waste runoff heavily threaten these clean water sources. By deploying IoT-based water quality sensors and predictive data analytics, we can monitor pollution levels, track resource degradation in real time, and ensure that local development does not destroy vital natural ecosystems.